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Beyond Sovereignty as Authority: The Multiplicity of European Approaches to Digital Sovereignty

Daniel Lambach & Linda Monsees

Abstract

Digital Sovereignty is core to many contemporary debates on the regulation of digital technology, securing supply chains and strengthening the digital economy. In this paper, we focus on the European debates and policies around this project. This paper argues that "digital sovereignty" is an attempt to rearticulate a European position in the ensuing Sino-American "tech war". This strategy is mainly symbolic and performative, but it also has some tangible outputs that make it more than mere rhetoric. We argue that the notion of digital sovereignty should be understood more as an overarching economic, societal and geopolitical project, rather than a project aimed at achieving any kind of political sovereignty or autarky in the digital sector. We arrive at this conclusion by analysing European policy documents and commentaries from various stakeholders for two cases: platform governance and semiconductors. The case of platform governance shows how the EU and its member states struggle to extend their regulatory power over the "data monopolies" of the major Silicon Valley companies. The other is semiconductors, where the EU has kickstarted several projects to improve the competitiveness of European manufacturers in a highly integrated global market. These examples demonstrate two things: a) that the pursuit of digital sovereignty does have some impact on the internet (platform regulation) but also extends to other technological fields beyond it (semiconductors); b) that digital sovereignty is not only a regulatory or technological project but also an economic, societal and geopolitical one.

Keywords

Internet fragmentation; digital sovereignty; international politics; platform governance; semiconductors

Word count: 8,939

Introduction

This paper speaks to the continuous debate on the notion of "digital sovereignty" (DS) and its role for EU policies. Among political scientists the idea of digital sovereignty has recently attracted attention as a focus of debates about sovereignty over and through networked technology. In that sense the debate relates to previous debates concerning the question of the control of cyberspace (Nye 2014; Eriksson and Giacomello 2006). DS is imbued with a range of different meanings ranging from the idea of "national internets" to more limited claims about industrial policy and legal liabilities of platform companies (Floridi 2020; Pohle and Thiel 2020; Bellanova, Carrapico, and Duez 2022). Common denominators are ideas about the legal regulation of digital spaces and strengthening technological autonomy. However, in this contribution we argue that the concept of digital sovereignty and the ideas it draws from have four dimensions. Not only is DS a techno-political project concerning its *regulation* but also in *economic*, *societal* and *geopolitical* terms.

In this paper, we offer a European viewpoint, specifically on what the European Union (EU) is doing when it refers to digital sovereignty. Our aim is neither to defend or justify EU discourses and strategies but to inject some nuance into the debate and contextualize European debates. We think that European discourse and practices of digital sovereignty exhibit elements from all four dimensions of DS: the regulatory, economic, social, and geopolitical one. Furthermore, we argue that there is no unified concept of sovereignty behind the rhetoric but rather a bricolage of different ideas.

Proposals for European digital sovereignty have certainly gained prominence within the EU and its member states. The European Commission in particular has argued that Europe needs to defend, regain, or expand its capacity and autonomy in the digital sphere. This – by itself – is not a fundamental challenge to the internet as we know it nor is it limited to classical notions

of political sovereignty as authority. First, European digital sovereignty designs are considerably less radical than alternative, more wide-ranging conceptions such as those espoused by China or Russia. Second, calls for digital sovereignty do not translate into a cohesive strategy but rather serve to organize diffuse political coalitions behind a vague political ideal. In this paper we show how the idea of digital sovereignty is worked through different projects in practice. This will illustrate how digital sovereignty mobilise diffuse political coalitions and works on a regulatory, economic, social and geopolitical dimension.

The sovereignty that European DS discourses reference is about more than just authority and power, infrequent references to Jean Bodin in academic treatises notwithstanding (e.g. in Mueller 2020, Pohle and Thiel 2020). While a sovereign ideal of the state being the final legal authority over digital matters features in parts of the debate, this is only one concern among several, and possibly not even the most prominent in European DS discourses. In the economic dimension, proponents of digital sovereignty echo mercantilist understandings of sovereignty by arguing that European digital industries need to become more globally competitive and call for appropriate industrial, innovation and research policies (Lambach and Oppermann 2022). As for the social dimension, digital sovereignty is supposed to protect European citizens' rights, their identities, democratic systems, and way of life. In the geopolitical dimension, digital sovereignty is considered a strategy to protect the EU and its values against rising geopolitical threats and technological dependencies, especially vis-à-vis China. Economic and societal ideas, we argue, have not received enough analytical attention in debates which are dominated by concerns about the possibilities and dangers of the idea of sovereignty.

The aim of this article is to illustrate the multiplicity of European thinking about digital sovereignty. We first detail how the European Union frames the problem that digital sovereignty is supposed to provide an answer for. We then demonstrate how DS works through the four dimensions of regulation, economy, society, and geopolitics by focusing on two very

different initiatives associated with DS. The example of platform governance highlights how DS is framed in terms of regulatory capacity, economic competitiveness, and social order. The semiconductor industry and the debates revolving around it show how DS is translated into an economic and social project that is shot through with references to geopolitical concerns. We conclude by arguing that DS is not only, or even primarily about legal authority but needs to be understood as something that is worked through a variety of practices and projects which are informed by a mix of ideas.

The European Union's Quest for Digital Sovereignty

Over the past decades digital technologies have become an integral part of the economy and society. But how should European states, societies, business, and citizens respond to the challenges of digitalisation? This large-scale transformation of societies and economies through the emergence of a "platform economy" or an "internet-industrial complex" (Flyverbom et al 2019) is worrying policymakers, commentators, and publics. European digital industries are seen as lagging their American and Chinese competitors. US-built platforms have created oligopolies which threaten the security of European user data. European attempts to regulate these platforms and support home-grown digital industries have had little impact. The concept of "digital sovereignty" is an attempt to navigate a course through this bundle of challenges.[1] Arguments for digital sovereignty typically hinge on safeguarding the self-determination of European societies in the digital age, to make digital industries globally competitive and to protect citizens from crime, surveillance, and the unauthorized use of their data (Sahin and Barker 2021: 11, Mueller 2020, Zeng et al 2017).

[1] In this paper we focus on the EU level but the concept as such was first discussed on the national level in France and Germany (cf. Lambach and Oppermann 2022, Bellanova et al. 2022: 346). For instance, the discussion about "la souveraineté numérique" (Bellanger 2014) in France can be traced back to 2012, with antecedents going back to the mid-2000s (Glasze et al 2022).

The EU is not alone in using the term. Chinese and Russian government officials used to speak of "internet sovereignty", "cybersovereignty", and "national Internet segments" but have more recently also started to frame their policies in terms of digital sovereignty (Drake et al 2016: 46, Jiang 2010, Litvinenko 2021, Zeng et al 2017). The term has by now spread across the globe (Chander and Sun 2021, Fischer 2022). For some outside of the United States, digital sovereignty represents resistance against "digital colonialism" or "data colonialism" by big platforms (Schneider 2022). In that sense, digital sovereignty discourses are a contemporary iteration of recurring debates about internet governance that go back to the late 1990s (Chenou 2014).

In Europe, the idea has been picked up by the European Union, especially under the von der Leyen Commission. This is driven by a perceived "need to preserve the European DNA of values and rights" (Celeste 2021: 212), and the term "has become increasingly popular among the EU's main institutional actors" (Roberts et al 2021: 2). Another explanation might be that DS creates inroads for the European Commission and other bureaucratic actors to discuss security issues, an area where member states jealously guard decision-making competencies.[2] However, as Roberts et al. argue, the term "lacks a clear definition and is deployed inconsistently" (Roberts et al 2021: 2). This echoes similar findings by Lambach and Oppermann (2022) who distinguish seven different narratives of digital sovereignty in German domestic politics. Germany has been an influential voice in European debates, especially as of late. The 2020 German presidency of the European Council aimed "to establish digital sovereignty as a leitmotiv of European digital policy" (Bundesregierung 2020: 8, see also Floridi 2020). While it is not the only, or even the first European country to discuss digital sovereignty, Germany stands out for making a strong and sustained case for it on the European level, putting its weight behind the concept in the EU.

[2] We thank one of the anonymous reviewers for this suggestion.

Previous scholarship on European approaches to digital sovereignty has been mostly policy-oriented and prescriptive, discussing whether and how the EU and its member states should pursue digital sovereignty (e.g. Burwell and Propp 2020, Christakis 2020, Hobbs 2020). Several contributions have also situated digital sovereignty in debates about sovereignty in political theory and legal philosophy (Floridi 2020, Pohle and Thiel 2020, Rone 2021), noting how uncertainties about the term sovereignty are mirrored in the conceptual confusion surrounding digital sovereignty. Celeste points out that digital sovereignty is only an iteration of debates that try to apply "the notion of sovereignty to the technological world" (Celeste 2021: 216). However, there has been little empirical research on the politics of digital sovereignty and on how the concept plays out in political discourse and practice. The few notable exceptions have traced portrayals of sovereignty in the digital sphere (Couture and Toupin 2019) or mapped different understandings of sovereignty onto agents, contexts, and values (Hummel et al 2021), albeit in very general terms.

Celeste (2021) and Roberts et al. (2021) provide more detailed accounts of European digital sovereignty debates, while other works look at discussions in national contexts (e.g. Lambach and Oppermann 2022). These works agree that digital sovereignty is a malleable concept that can be used to justify a multiplicity of policies. Roberts et al. (2021) identify five different, if related policy areas, in which the EU speaks of digital sovereignty, and participants in the debate often switch back and forth between meanings to suit their needs (Ruohonen 2021: 441). Lambach and Oppermann argue that this conceptual multifariousness of digital sovereignty is "not a bug but a feature" (Lambach and Oppermann 2022: 13) because it allows disparate actors to organize political coalitions around a shared, if ill-defined notion.

Building on this scholarship, we distinguish between four different ideas or dimensions of European DS discourse and practice: *the regulatory*, *the economic*, *the societal* and *the geopolitical* (security) one, which we explain below. The first one has certainly attracted the

most attention and is the most visible one: digital sovereignty concerns global networked technology and questions about its usage and regulation. The last dimension concerning geopolitics becomes visible when we see how justifications about DS are entangled with narratives about rising geopolitical rivalries (see [Anon] 2022). The idea of DS as a project concerning the economy and society has received less attention. However, in the empirical section we will show their importance. To be clear, the distinction between those four ideas is purely analytical. Empirically, they are all closely entangled and cannot be neatly separated. These ideas are not specific narratives (in the sense the term is used by Lambach and Oppermann 2022) but exist in the background. In this sense, they are more like imaginaries about the purpose and meaning of sovereignty in a digital and interconnected world. Identifying, distinguishing and examining these four ideas is helpful in clarifying the terms of the debate but also examine where critiques of DS might target the 'wrong' ideas and assumptions. Ultimately, we argue that the idea of sovereignty in a legal and regulatory sense is given too much credit while the social and economic dimension are *de facto* more important for EU policy debates than academic commentary acknowledges.

The idea of digital sovereignty as legal authority and regulatory power over networked technology is associated with classical understandings of sovereignty in political and legal theory. Notions of self-determination, autonomy, and freedom from external authority are central to this understanding, while other aspects of sovereignty – the capacity to enforce laws within a circumscribed territory as well as concepts of *popular sovereignty* which locate the source of sovereignty and ultimately its legitimacy within the populace – are less prominent (Pohle and Thiel 2019, Rone 2021). Applied to the digital sphere, this implies an extension of state control over what is considered by some as an autonomous space (Lambach 2020). Milton Mueller, in particular, has been highly critical of this line of thinking. He argues that attempts to impose state authority over the internet are incompatible with the network character of the

technological infrastructure (Mueller 2020). On a more practical level, imposing legal authority over internet activities has proven to be challenging. European attempts to implement laws, e.g. the Right to be Forgotten, the "Cookie Directive" or the German *Netzwerkdurchsetzungsgesetz* (Network Enforcement Act) have all led to protracted conflicts with non-European platforms. The problem of *where* to apply these laws and the resulting emergence of extraterritorial forms of regulation are still unsolved and themselves represent sources of friction in internet governance and international politics more broadly. Practical problems aside, the idea to extend regulatory power over (global and networked) digital technology is still prominent within DS.

The second idea of digital sovereignty as economic competitiveness is related to the first. Although its principal concerns are different, it draws from the first a hope that European-based platforms would be easier to regulate than overseas ones. Beyond that, this idea is mainly concerned with maintaining the innovative potential of European economies. Digitalization, it is often asserted, is challenging the business models of European manufacturing industries, which represent the core engine of prosperity for the continent (Bundesverband der deutschen Industrie 2020, European Institute of Innovation & Technology 2020). To be sure, this is nothing new – worries about a "technology gap" between Europe and the United States and, for a time, Europe and Japan go back to the 1950s (Peterson and Sharp 1998: 4-9). Nowadays, Silicon Valley start-up culture and its high-risk, high-reward mantra of "move fast and break things" is seen as the gold standard of digital innovation but is difficult to replicate in what remains of Rhenish capitalism (van Dijck 2020). Hence, digital sovereignty is about making European companies competitive in a changing global economy. This is a steep challenge. American tech companies such as Google, Apple, Amazon, or Intel as well as their Chinese counterparts (Alibaba, Tencent, Huawei etc.) are often quasi-monopolistic providers of high-tech products and digital services (Atal 2021).

This issue goes beyond platforms, as these businesses have come to be known (Gillespie 2010, Poell et al 2019). Markets for hardware manufacturing, like 5G mobile communications or high-end semiconductors, and digital products not aimed at consumers, like Artificial Intelligence, are also dominated by small numbers of big players. In some digital markets, especially platforms, network effects and economies of scale create oligopolies; in others, such as semiconductors, prohibitive entry costs entrench technological leadership. Competitive advantages in one sector translate into advantages in other sectors. For instance, the user data gathered by platform companies in their day-to-day operations is very valuable for the development and training of machine learning models, and the "cloud" business was basically created by Amazon renting out excess server capacities (Striphas 2015: 396). Given these market conditions, European attempts have been less focused than in the past on nurturing competitive European champions (the "Airbus model"). Instead, one strategy has been to work with the "internet giants" to make sure that certain value-adding industries are located in Europe (see the example of Intel below). Another has been to make international players conform to European regulations, e.g., through the GAIA-X cloud infrastructure project, and to thereby open spaces for European companies to come up with new business models alongside the market heavyweights (Bendiek and Stürzer 2022).

The third idea is that Europe needs digital sovereignty to protect its citizens, social fabric, and democratic systems (Kagermann and Wilhelm 2020). European countries have different approaches to civic rights and civil liberties compared to the United States, most notably in areas like freedom of speech or data protection. Having to follow US practices in these regards and users being subjected to what would be considered harmful speech in Europe is seen as injurious to these rights. Hate speech and fake news, spread mostly unchecked through US platforms, are seen as fuelling conspiracy theories, and playing into the hands of anti-democratic forces (van Dijck 2020). European citizens, the proposals claim, need better

protection and better digital literacy skills to become more "digitally sovereign" users of these services (Gesellschaft für Informatik 2020). An undercurrent of culturalism is visible in these discussions, where the supposedly more ethical, European "Us" is contrasted with a soulless, neoliberal Silicon Valley "Other" (van Dijck 2020).

The fourth and final idea concerns geopolitics and security. The idea of DS is strongly interlinked with the idea of rising geopolitical rivalry (Guzzini 2013) and assumes that strengthening DS will either help to protect against foreign attacks of espionage or that it will help to secure supply chains of important technology ([Anon] 2022). As we will discuss in more detail below, securing semiconductor supply chains is a cornerstone of DS policies and explicitly justified with geopolitical rivalries. Economic dependencies on political rivals and enemies need to be reduced (Alam et al. 2020; Cartwright 2020). DS is thus not only a legislative or infrastructural project but also tightly linked to security concerns. In the case of the EU, the notion of security is a broad one where ideas of societal and economic security go hand in hand with national security (Nyman and Burke 2016; Christou et al. 2010). The geopolitical dimension is strongly interlinked with the previous three ideas. Strengthening innovation and securing societal values are also considered direct security concerns. That is why we consider these four ideas to be analytically separate but empirically entangled. For example, to sanction harmful speech online (a societal concern), authorities need jurisdictional competencies over the platform (a regulatory concern), a task which is made easier if the company running the platform is actually situated in Europe (an economic concern).

In the rest of the paper, we explore these four dimensions in more detail and show how they are interlinked. Of course, not all four ideas might be present in every debate or project related to DS, but we show below how different ideas and justifications of DS come to the fore. We explore this notion through two case studies. The first one is about platform governance in the EU and is focused on the regulatory and societal ideas informing DS debates therein. Thesecond

one is about the European semiconductor industry and highlights how digital sovereignty is mainly presented in economic and geopolitical terms. These two examples illustrate the breadth of the European debate about the topic. This also means that our empirical material stems from the EU level and consists mainly of official policy documents, public speeches, think tank publications, and press reports. Even though debates on both issues are also conducted on national levels, what interests us here is how the debate occurs within EU institutions and what ideas it reflects.

European Platform Governance

Platform governance is one of the core concerns of the European Union's pursuit of digital sovereignty. Platforms, or "intermediary services", as they are referred to in EU parlance such as the Digital Services Act (European Commission 2020), "are increasingly seen as dominating entire sectors of the EU economy and depriving EU Member States of their sovereignty in areas such as copyright, data protection, taxation or transportation" (Madiega 2020: 4). This perception is the result of two intertwined trends: regulatory evasion by big internet companies, and the oligopolization of digital markets.

The first point draws on uncertainty about the nature of platforms. In terms of their function, platforms are online "architectures" (Popiel 2022: 29) or "infrastructures" (Poell et al 2019: 3) that organize relations and interactions between end-users and services (the "ecosystem" part of the platform) as well as among end-users themselves (the "social media" part). Platform companies "have argued they are mere 'facilitators', connecting users to creators or producers, and connecting content to users; insisting on their status as 'connectors' and avoiding regular legal categories, platforms and their operators have avoided taking responsibility" (van Dijck 2020: 2). This is the product of an active self-construction to evade existing regulatory concepts. As Gillespie argued in his seminal article, "these firms are working not just politically but also

discursively to frame their services and technologies […] to strike a regulatory sweet spot between legislative protections that benefit them and obligations that do not" (Gillespie 2010: 348). This is key to the platform business model: companies do not want to be held liable for the services they offer. YouTube claims it is not liable for the videos it hosts, same for Facebook and user comments, and for Uber and the working conditions of its drivers.

Secondly, the emergence of platforms has also changed digital markets, which show strong tendencies towards concentration, creating oligopolies or monopolies for key services. It has arguably even changed the nature of the internet itself. Driven by the shift to user-generated content in the "web 2.0" in the early 2000s, the internet evolved "towards a proprietary and concentrated platform economy" (Bietti 2022: 3) where users congregate in a shrinking number of corporate "walled gardens" (Lambach 2020: 498-99). The reasons for these are manifold, such as the network effects that make popular social media sites particularly attractive for new customers and advertisers. Big firms can leverage the data they gather into more effective advertising and consumer behavior research or use it as training data for machine learning algorithms. Platforms can also leverage their dominant position in one market for advantages in other markets through "bundling" practices, e.g., like Microsoft exploited its monopoly in PC operating systems (Windows) to induce customers to use its web browser (Internet Explorer) until US and EU authorities stepped in. Platforms also defend their position by buying emerging companies either to head off potential competition or to expand their business into other markets. As Pasquale argues, timing and luck mattered for firms like Alphabet, Meta, Amazon or Microsoft to emerge as the dominant players in their respective markets: "The first platform to gain critical mass can leverage that advantage into massive financing, which in turn can scare away competitors" (Pasquale 2016: 316). In short, platforms distort markets by becoming markets themselves (Bietti 2022).

For the European Union, the market-distorting power of platforms would already be a challenge by itself. Although the EU and member states have well-established regulatory institutions and robust systems of competition law, dealing with "platforms" that consciously evade regulatory categories, is very difficult. But the problem is further exacerbated by the fact that the platforms dominating European business are located in the United States. Measured in terms of market capitalization, Alphabet, Meta, Microsoft, Amazon, and Apple are the largest platforms worldwide. Even "second-tier" platforms like Netflix, Salesforce or Paypal are larger than the largest European digital company, SAP (Economist 2020).[3] Europe is seen as "dependent on the American platform ecosystem" (van Dijck 2020: 2), with some voices, especially in France, even speaking of "digital colonialism" by the United States (Christakis 2020: 6-7, Miailhe 2018).

The question of what to do about these foreign internet giants has loomed large in European politics for the past decade or two. The idea to mitigate deleterious effects on employment standards, democratic stability, and civil rights is seen as a cornerstone of digital sovereignty. The European Commission attempts to regulate the dominant players through what has become known as "the Brussels effect", i.e. the extraterritorial application of EU laws (Bradford 2020).[4] This necessitated a shift in how the EU approaches these companies, which enjoyed substantial liability indemnities until the early 2010s. It was only after European courts and regulatory bodies began to treat platforms as business entities *sui generis* that they started holding these companies to account (Katzenbach 2021: 8-9).

[3] Globally, Asian companies like Alibaba or Tencent are the closest competitors for the U.S. "Big Five". Due to the limited presence of Chinese firms in Europe, these are not the platforms that European digital sovereignty debates are directed against.

[4] Besides the extraterritorial application of EU laws, the diffusion of European regulations to other countries can be considered another example of the Brussels effect. For example, the UK government in May 2022 publicised plans to regulate tech firms that are strongly reminiscent of the Digital Markets Act. See https://www.gov.uk/government/news/government-sets-out-plans-for-how-tech-regulator-will-tackle-dominance-of-major-firms.

Platform regulation started with piecemeal efforts that addressed particular aspects of digital business, such as the e-commerce directive (2000), which first transposed the principles of the internal market to online services, and the ePrivacy directive (2002). Since then, a variety of regulations has accreted, many of which updated earlier directives and regulations, most notably the General Data Protection Regulation (GDPR, 2016), which had substantial extraterritorial effects. To streamline, update, and expand upon this system of regulations, the Commission has been working on several pieces of legislation since 2020 (Bendiek and Stürzer 2022): the Digital Markets Act (DMA) which was enacted on 14 September 2022, the Digital Services Act (DSA) entered into force on 16 November 2022, , the Data Governance Act (DGA) which was published on 30 May 2022, and the Data Act (DA) which is still at the proposal stage and expected to be finalized in 2023.

Of the four, the DMA and DSA form the heart of European platform regulation and address global platform companies most directly. The Digital Markets Act (DMA) identifies "gatekeepers", i.e. companies with entrenched advantages in digital markets, and limits the possibilities of these companies to stifle competition. Gatekeepers will be obliged to offer nondiscriminatory access to their marketplaces and devices to third-party vendors and to make messenger services interoperable. Fines up to ten percent of a firm's global turnover can be levied to ensure compliance, thereby expanding the GDPR's previous limit of four percent. The Digital Services Act (DSA) directly targets "very large online platforms" (VLOPs, Art. 25) and search engines, i.e., those with more than 45 million users in the EU, and places them under a special supervisory regime which obliges them, e.g., to create systems that deal with legal complaints and user reports within defined timeframes. While the DMA regulates markets, the DSA regulates digital products and digital content, replacing the e-commerce directive.

The DGA and DA are cornerstone proposals of the EU Data Strategy, which aims at creating a "single market for data" and "to make the EU a leader in a data-driven society".[5] Together, they provide rules for data-sharing to make new business models viable. The aim is to open up commercial possibilities while balancing these with data protection concerns enshrined in the GDPR. In contrast to the DMA and DSA, these two acts only indirectly address platforms, often via reference to the DMA. For instance, the DA provides a mechanism for data sharing that makes it easier for users to access and use their personal data collected by digital services and platforms. This includes sharing this data with third parties, a category from which gatekeepers (using the DMA criteria) are explicitly excluded (DA, Art. 5.2).

While the practical impact of these acts is yet to be seen, we can assess the ideas that drove the development of the proposals. All four legislative items are explicitly connected to the pursuit of digital sovereignty. As the Commission frames its approach: "Europe must now strengthen its digital sovereignty and set standards, rather than following those of others"[6], while EC Vice-President Věra Jourová calls for a "digital rulebook" to foster trust and innovation, and for Europe to be "an agent, not an object" of the digital revolution.[7] The debate features references to three of the four ideas mentioned above: regulation, economic competitiveness, and social order.

Across all four acts, there is a strong element of establishing or clarifying legal regulation. The aim of the DSA is to define "clear responsibilities and accountability for providers of intermediary services, and in particular online platforms, such as social media and marketplaces" (DSA, p. 2). The regulatory arsenal will be expanded, and the extraterritorial application of the acts is clearly intended. Big Tech firms that do not comply with the DMA can expect large fines and may even risk being forcibly broken up. Companies will be obliged

[5] https://ec.europa.eu/info/strategy/priorities-2019-2024/europe-fit-digital-age/european-data-strategy_en
[6] https://ec.europa.eu/info/strategy/priorities-2019-2024/europe-fit-digital-age_en
[7] https://ec.europa.eu/commission/presscorner/detail/en/SPEECH_22_851

to give access to data, including trade secrets, to "monitor and assess compliance" (Art. 31) with the DSA. However, platforms still retain certain privileges. For example, even under the DSA platforms cannot be held liable for illegal speech if they are unaware of it. But while the DSA establishes a regulatory regime for VLOPs, the regime only infringes upon the commercial freedoms of platforms in a limited way, preserving most of their self-governance.

The *economic ideas* of improving the competitiveness of European digital industries are also clearly visible. All four Acts view "big tech companies as bottlenecks that need regulating to let other smaller businesses enter digital market segments and compete with incumbents" (Bietti 2022: 66). Furthermore, lock-in and network effects create market failures, with entrenched monopolists engaging in "unfair practices" which lead to "inefficient outcomes in the digital sector in terms of higher prices, lower quality, as well as less choice and innovation to the detriment of European consumers" (DMA, p. 1). This stifles competition and limits the innovative potential of European companies, whether "(s)tart-ups, small and medium-sized enterprises and companies from traditional sectors with less-developed digital capabilities" (DA, p. 25). European regulation therefore aims at re-establishing fairness and competition in digital markets (European Commission 2022a), notably through establishing a "common European data space, a Single Market for data" (DGA, p. 9), and at fostering innovation in European industries.

Regulating platforms is also seen as maintaining and/or improving *social order*. The DSA in particular is very explicit that its purpose is to safeguard legal protections on online platforms for end users and to mitigate these platforms' potential risks for democracy and social stability. Art. 26 obliges VLOPs to report annually on "any significant systemic risks stemming from the functioning and use made of their services" including "any negative effects for the exercise of […] fundamental rights" (Art. 26, 1(b)) and "manipulation of their service […] with an actual or foreseeable negative effect on the protection of public health, minors, civic discourse, or

actual or foreseeable effects related to electoral processes and public security" (Art. 26, 1(c)). Conversely, the DGA presents the opportunities that arise from common European data spaces in sectors such as "health, environment, energy, agriculture, mobility, finance, manufacturing, public administration and skills"[8]. This kind of "data-driven innovation will bring enormous benefits for citizens, for example through improved personalised medicine, new mobility, and its contribution to the European Green Deal" (DGA, p. 9). In short, EU platform governance is informed by three of the four ideas identified above, with legal regulation being positioned as an instrument that also improves economic competitiveness and social order.

The European Semiconductor Industry

Without the invention of semiconductors, computing the way we know it today would be impossible. Semiconductors are necessary for all digital devices from car keys to mobile phones and are a key component of hardware innovation. High-end semiconductors which are needed for the development of complex machine-learning technology are of special importance (Ezell 2020: 2).[9] All semiconductors are produced in highly specialised global supply chains and since the 1960s the semiconductor industry has grown almost continuously (Mathews and Dong-Sung 2000: 33). East Asian countries had identified semiconductors as a strategically important industry early on: First, Japan entered the market, closely followed by South Korea and Taiwan (Mathews and Dong-Sung 2000: 30). Nowadays, the production of semiconductors relies on a global, interconnected supply chain – even the US and China are dependent on other countries and far away from being able to develop, produce and implement high-end chips on their own.

Semiconductors are, and always have been, a political technology (Pearson and Brattain 1955). The ongoing chip shortage resulting from the Covid-19 pandemic is only the latest incident

[8] https://digital-strategy.ec.europa.eu/en/policies/data-governance-act

[9] Despite this focus on high-end chips, the role of "larger" or less strong semiconductors cannot be emphasized enough. They are necessary for virtually all infrastructures in the EU.

where semiconductors became a global "public problem" (Dewey 1999, Miller, Keohane and Inagaki 2021). Because production plants in China were closed and shipments disrupted, chips were (and still are) lacking in multiple industries. Due to their importance for both everyday technological devices and cutting-edge innovation, semiconductors are a concern for (economic) policies around the world and they are especially important for relations among the EU, China and the US ([Anon] 2022). Geopolitical tensions between these three powers shape policies about semiconductors since dependencies on other nations is increasingly seen as a problem. However, as this article argues, geopolitics are not the only idea that is driving semiconductor politics. Indeed, when we look closer at current EU policies, we can identify a much broader approach towards the role of semiconductors.

In the EU, semiconductors are part of the wider DS strategy. They are mentioned in various DS papers as an example of where the EU needs to become more active and needs to invest more over the next years (Kleinhans 2021: 6). Directly linked to digital sovereignty, the EU initiative for supporting semiconductors and cloud technology attempts to "[join] forces to maintain and boost the competitiveness in the sector in the EU" (European Commission 2021). A central cornerstone of this initiative is the Alliance on Processors and Semiconductors aimed at "the reinforcement of the European electronics design ecosystem" and "the establishment of the necessary manufacturing capacity" (European Commission 2022b). In the related declaration the signing member states frame the Alliance's purpose as "reinforc[ing] the processor and semiconductor ecosystem and to expand industrial presence across the supply chain, in order to address key *technological, security and societal challenges*" (European Union 2020, emphasis added). Indeed, the aim is to avoid "excessive dependencies on imports" (European Union 2020). This shows how the EU considers semiconductors as a security technology in the sense that their availability is necessary for a functioning digital economy. This of course needs to be understood in a wider context in which the EU combines security with economic policies. We

can thereby see how the economic and geopolitical dimension are entangled through the notion ofDS. Semiconductors as part of the EU DS strategy are thus not only a purely technical project but also bring in ideas about economic competitiveness and geopolitical concerns. Covid-19 and the related disruption of supply chains and chips shortages especially in the automotive industry functioned as a catalyst for these policies.

As part of their DS strategy the Commission has proposed a "Chips Act" which is currently in preparation. Commission President Ursula von der Leyen summarized the Act's aim as follows: "we are combining investment, a regulatory framework and the necessary strategic partnerships to make Europe a leader in this market that is so important […] chips are at the centre of the global technological race [and a] bedrock of our modern economies" (von der Leyen 2022). Part of the Chips Act is also to attract investment in this area (European Commission 2022c). These policies thus encompass mechanisms to increase research and development as well as supporting industries in their production of higher end semiconductors and computer chips. It has recently emerged that US technology giant Intel will be investing 33 billion Euros in Europe, the main investment being a semiconductor fabrication plant ("fab") in Magdeburg, Germany, whose construction is scheduled to start in 2023. There was intense competition among different locations within Europe for this plant, highlighting the importance of these fabs for national and local economies (Goodkind 2021). So, while on the one hand Intel's plans were seen through a geopolitical DS lens, as a strengthening of European capacities in a strategic technology sector, the debate was also about the economic benefits this plant would have for the regions competing for it. Here we can see how this DS project gets translated into more "traditional" economic policies and is embedded in debates about the future of the European (and German) economy.

Semiconductors are not only a hugely important technology but also, rather more mundanely, an industry that requires subsidies to attract global investment. This point also comes to the fore

when we look at the *de facto* production process. Because the production of semiconductors is reliant on complex and intricate supply chains any kind of autonomy (much less autarky) which terms like "digital sovereignty" or "technological sovereignty" promise is simply not possible. In addition, semiconductors rely on high up-front investment in research and development, making market entry difficult (Kleinhans and Baisakova 2020). Furthermore, the capabilities of semiconductors are improving quickly, and the chips are becoming ever smaller, creating an immensely steep curve for businesses trying to catch up to their more established competitors. Therefore, only very few companies produce high-end semiconductors and single-country efforts to achieve autonomous production capacities in the high-end segment of the market are unlikely to be competitive.

As a result, any debate around "autonomy" or "sovereignty" in semiconductors is ambiguous. On the one hand the EU clearly uses the idea of "strategic autonomy" as a justification for supporting industries and strategic value chains (Strategic Forum 2019). But even at the highest political level it is by now acknowledged that autonomy in the sense of avoiding dependencies is impossible. For example, in von der Leyen's speech quoted above she is very explicit: "It should be clear that no country – and even no continent – can be entirely self-sufficient. This is impossible. Europe will always work to keep global markets open and to keep them connected" (von der Leyen 2022). This shows that the aim of these policies is by now more about managing dependencies and increasing resilience.

In line with the overall aim of the paper, we can also identify a societal dimension in the debates around semiconductors. Ideas about DS as a societal project enter the debate in the form of arguments about how (better) semiconductors will allow for a future-oriented and sustainable society. In the declaration on semiconductors, it is stated that semiconductors are crucial to find answers to "key technological, security and societal challenges" (European Union 2020). This is because semiconductors "determine the characteristics of the products into which they are

embedded – including security, privacy, energy performance and safety – shaping how Europe's green and digital transition will unfold" (European Union 2020). Semiconductors play a central role in a vision of the future of the EU that is sustainable, protects citizen rights' and allows for a safe and secure society. This again shows how the notion of "sovereignty" is used in ways that do not refer to the idea of political authority. Digital sovereignty in the context of semiconductors is mainly used to promote a certain vision of the EU's digital economy and society.

To conclude, semiconductors are a crucial technology and the political debate about it is shot through with geopolitical remarks. However, we have shown in this section how DS policies are not primarily about achieving regulatory sovereignty or economic autonomy. Notion of sovereignty-as-authority are not feasible when acknowledging the transnational production process and fast pace of development. Talking about digital sovereignty in the case of semiconductors thus cannot mean that the EU will be able to produce all components of high-end semiconductors on their own while at the same time being competitive on the global market. Semiconductor policies are primarily economic policies aimed at strengthening the EU vis-à-vis its competitors. In addition, the DS debate on semiconductor taps also into ideas about societal progress enabled by better semiconductors and of course notions of security. Thus, we can see how different ideas about DS are all entangled and combined into a single argument of why strong semiconductor industries and supply chains are important for the future of the EU.

Conclusion

The two examples discussed above demonstrate the multiplicity of the European DS agenda. Platform regulation and the semiconductor strategy address very different challenges, involve different actors, and use different instruments. In short, if an issue involves digital technology, whether hardware, software, or data, it is liable to be drawn into this discussion and reframed

as an aspect of digital sovereignty. As a result, the DS agenda of the European Union is fairly unfocused. However, we find that there are some consistent concerns that drive EU positions in these areas and political engagement with these issues. The first is that the EU and member states look for digital sovereignty to establish legal authority over digital markets. Second, the EU aims for digital sovereignty to improve the economic competitiveness of digital and traditional industries. Third, the EU wants digital sovereignty to maintain social and democratic stability. And fourth, DS is part of a broader geopolitical debate on the role of the EU in global politics. These four ideas come up in different combinations in various policy fields. Platform regulation debates are especially shaped by regulatory, economic, and societal arguments, whereas the semiconductors discussion focuses mainly on the economic side but also combines strong geopolitical concerns with some ideas about a future EU society. Clearly, there is more going on underneath the umbrella of "sovereignty" than Bodin-style political theory would lead us to expect.

We need to look to the sources of these ideas and concerns. The European Union's concern with digital sovereignty reflects both a long-term insecurity as well as current international discourses. The term "sovereignty" is particularly captivating in a geopolitical and economic environment that is perceived as uncertain, rapidly changing and threatening. It is hard not to see contemporary debates as the most recent reiteration of classic fears of a "technology gap" vis-à-vis more advanced global competitors. The EU is perenially concerned about its economic competitiveness and innovative capabilities. Digital technologies are presented as a field of struggle in the unfolding geopolitical struggle between the United States and China or – in the extended version – between democracy and autocracy. These "tech war" narratives lead to a geopoliticization of technology that ramps up the salience of long-running "tech gap" assessments (Christakis 2020, Sahin and Barker 2021). In addition, DS debates draw on global discussions about digital technology, specifically "technology as geopolitical competition" and

"technology as a threat to democracy" narratives. In this context digital media is then also portrayed

as a set of "echo chambers" full of "fake news" that threaten democratic quality and stability. It draws on the generalized feeling that digital technologies destabilize political systems, whether they are democratic or autocratic. Digital sovereignty then conveys a notion of stability and control and thus works particularly well in these political debates.[10]

Debating digital sovereignty also points us to larger questions concerning internet fragmentation and the regulation of global networked technology. So far, we expect the impact of European DS projects on internet integrity to be quite limited. To be sure, platform regulation will have effects through the extraterritorial application of EU standards and limit commercial freedoms of platform operators, leading to either a globalization of EU regulations or territorially differentiated services offered by global companies. Of course, this assumes that the regulations will actually be applied as written, which is far from certain, as competition law and internet regulation have been persistent sticking points in transatlantic relations (Farrell 2003). Furthermore, there are aspects of the DS agenda – semiconductors are a case in point – that have little to no influence on internet integrity and fragmentation. Even if some kind of "autonomy" is aspired to in political rhetoric, this seems unachievable for technical reasons and would even be disadvantageous. Digital sovereignty as a concept will thus likely stick around for a bit longer even though its effect might be less about enforcing "sovereignty" in a regulatory sense but rather about pushing ideas about the future of EU security, economy and society.

References

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Alam, S., Chu, I., Lohokare, S. and Saito, S. (2020) *Globality and Complexity of the Semiconductor Ecosystem*. Accenture/GSA. https://www.gsaglobal.org/globality-and-complexity-of-the-semiconductor-ecosystem/.

Atal, M. R. (2021) 'The Janus faces of Silicon Valley', *Review of International Political Economy*, 28(2): 336-50.

Bellanger, P. (2014) *La souveraineté numérique,* Paris: Éditions Stock.

Bellanova, R., Carrapico, H., and Duez D. (2022) 'Digital/Sovereignty and European Security Integration: An Introduction'. *European Security* 31(3): 337–55.

Bendiek, A. and Stürzer, I. (2022) *Advancing European Internal and External Digital Sovereignty: The Brussels Effect and the EU-US Trade and Technology Council*, Berlin: German Institute for International and Security Affairs.

Bietti, E. (2022), A Genealogy of Digital Platform Regulation. SSRN. http://dx.doi.org/10.2139/ssrn.3859487

Bradford, A. (2020) *The Brussels Effect: How the European Union Rules the World*, Oxford: Oxford University Press.

Bundesregierung (2020) *Together for Europe's recovery: Programme for Germany's Presidency of the Council of the European Union (1 July to 31 December 2020)*: Council of the European Union.

Bundesverband der deutschen Industrie (2020) *Europas digitale Souveränität nachhaltig stärken*, Berlin: Bundesverband der deutschen Industrie.

Burwell, F. G. and Propp, K. (2020) *The European Union and the Search for Digital Sovereignty: Building "Fortress Europe" or Preparing for a New World?*: Atlantic Council.

Cartwright, M. (2020) 'Internationalising State Power through the Internet: Google, Huawei and Geopolitical Struggle'. *Internet Policy Review*, 9(3).

Celeste, E. (2021) 'Digital Sovereignty in the EU: Challenges and Future Perspectives', in F. Fabbrini, E. Celeste and J. Quinn (eds), *Data Protection Beyond Borders: Transatlantic Perspectives on Extraterritoriality and Sovereignty*, Hart Publishing. pp. 211-28.

Chander, A. and Sun, H. (2021) *Sovereignty 2.0*: University of Hong Kong Faculty of Law Research Paper 2021/041.

Chenou, J.-M. (2014) 'From Cyber-Libertarianism to Neoliberalism: Internet Exceptionalism, Multi-stakeholderism, and the Institutionalisation of Internet Governance in the 1990s', *Globalizations*, 11(2): 205-23.

Christakis, T. (2020) *"European Digital Sovereignty": Successfully Navigating Between the "Brussels Effect" and Europe's Quest for Strategic Autonomy,* Grenoble: Multidisciplinary Institute on Artificial Intelligence/Grenoble Alpes Data Institute.

Christou, G., Croft, S., Ceccorulli, M. and Lucarelli S. (2010) 'European Union Security Governance: Putting the "Security" Back In', *European* Security 19(3): 341–59.

Couture, S. and Toupin, S. (2019) 'What does the notion of "sovereignty" mean when referring to the digital?', *New Media & Society*, 21(10): 2305-22.

Dewey, J. (1999) *The Public and Its Problems*, Athens: Swallow Press.

Drake, W. J., Cerf, V. G. and Kleinwächter, W. (2016) *Internet Fragmentation: An Overview*, Geneva: World Economic Forum.

Economist (2020): Who will benefit most from the data economy? *The Economist*, 20 February 2020. https://www.economist.com/special-report/2020/02/20/who-will-benefit-most-from-the-data-economy

Eriksson, J. and Giacomello, G. (2006) 'The Information Revolution, Security, and International Relations: (IR)Relevant Theory?', *International Political Science Review* 27(3): 221–44.

European Commission (2020) Proposal for a Regulation of the European Parliament and of the Council on a Single Market For Digital Services (Digital Services Act) and amending Directive 2000/31/EC. Brussels: European Commission, COM (2020) 825, 15 December 2020. https://eur-lex.europa.eu/legal-content/en/TXT/?qid=1608117147218&uri=COM%3A2020%3A825%3AFIN

European Commission (2021) Declaration of the Industrial Alliance for Processors and Semiconductor Technologies. https://digital-strategy.ec.europa.eu/en/policies/alliance-processors-and-semiconductor-technologies.

European Commission (2022a) Data Act: Commission proposes measures for a fair and innovative data economy. Press Release, 23 February 2022. https://ec.europa.eu/commission/presscorner/detail/en/ip_22_1113

European Commission (2022b) Alliance on Processors and Semiconductor Technologies | Shaping Europe's Digital Future. 2022. https://digital-strategy.ec.europa.eu/en/policies/alliance-processors-and-semiconductor-technologies

European Commission (2022c) European Chips Act: Factsheet | Shaping Europe's Digital Future. https://digital-strategy.ec.europa.eu/en/library/european-chips-act-factsheet

European Institute of Innovation & Technology (2020) *European Digital Infrastructure and Data Sovereignty. A Policy Perspective*.

European Union (2020) Joint declaration on processors and semiconductor technologies https://digital-strategy.ec.europa.eu/en/library/joint-declaration-processors-and-semiconductor-technologies

Ezell, Stephen (2020): *An Allied Approach to Semiconductor Leadership. Information Technology & Innovation Foundation*, Washington DC: Information Technology and Innovation Foundation. https://itif.org/publications/2020/09/17/allied-approach-semiconductor-leadership

Farrell, H. (2003) 'Constructing the International Foundations of E-Commerce: The EU-U.S. Safe Harbor Arrangement', *International Organization*, 57(2): 277-306.

Fischer, D. (2022) 'The digital sovereignty trick: why the sovereignty discourse fails to address the structural dependencies of digital capitalism in the global south', *Zeitschrift für Politikwissenschaft*.

Floridi, L. (2020) 'The Fight for Digital Sovereignty: What It Is, and Why It Matters, Especially for the EU', *Philosophy & Technology*, 33(3): 369-78.

Flyverbom, M., Deibert, R. and Matten, D. (2019) 'The Governance of Digital Technology, Big Data, and the Internet: New Roles and Responsibilities for Business', *Business & Society*, 58(1): 3-19.

Gesellschaft für Informatik (2020) *Schlüsselaspekte Digitaler Souveränität*.

Gillespie, T. (2010) 'The politics of 'platforms'', *New Media & Society*, 12(3): 347-64.

Glasze, G., Cattaruzza, A., Douzet, F., Dammann, F., Bertran, M.-G., Bômont, C., Braun, M., Danet, D., Desforges, A., Géry, A., Grumbach, S., Hummel, P., Limonier, K., Münßinger, M., Nicolai, F., Pétiniaud, L., Winkler, J. and Zanin, C. (2022) 'Contested Spatialities of Digital Sovereignty', *Geopolitics*.

Goodkind, N. (2021) Biden administration sounds the alarm on the semiconductor crisis. Fortune Magazine, 17 July 2021, https://fortune.com/2021/07/16/biden-administration-sounds-the-alarm-on-the-semiconductor-crisis/

Guzzini, S. (ed.) (2013) *The Return of Geopolitics in Europe? Social Mechanisms and Foreign Policy Identity Crises*, Cambridge: Cambridge University Press.

Hobbs, C. (2020) *Europe's Digital Sovereignty: From Rulemaker to Superpower in the Age of US-China Rivalry*, London: European Council on Foreign Relations.

Hummel, P., Braun, M., Tretter, M. and Dabrock, P. (2021) 'Data sovereignty: A review', *Big Data & Society*, 8(1).

Jiang, M. (2010) 'Authoritarian Informationalism: China's Approach to Internet Sovereignty', *SAIS Review of International Affairs*, 30(2): 71-89.

Kagermann, H. and Wilhelm, U. (2020), European Public Sphere: Gestaltung der digitalen Souveränität Europas. *acatech IMPULS*, München acatech.

Katzenbach, C. (2021) 'Die Governance sozialer Medien', in J.-H. Schmidt andM. Taddicken (eds), *Handbuch Soziale Medien*, Wiesbaden, Springer Fachmedien Wiesbaden.

Kleinhans, J.-P. and Baisakova, N. (2020) *The global semiconductor value chain; A technology primer for policy makers*, Berlin: Stiftung Neue Verantwortung. https://www.stiftung-nv.de/sites/default/files/the_global_semiconductor_value_chain.pdf

Kleinhans, J.-P. (2021): *The lack of semiconductor manufacturing in Europe. Why the 2nm fab is a bad investment*. Berlin: Stiftung Neue Verantwortung Policy Brief, March 2021.

Lambach, D. (2020) 'The Territorialization of Cyberspace', *International Studies Review*, 22(3): 482-506.

Lambach, D. and Oppermann, K. (2022) 'Narratives of digital sovereignty in German political discourse', *Governance*.

Litvinenko, A. (2021) 'Re-Defining Borders Online: Russia's Strategic Narrative on Internet Sovereignty', *Media and Communication*, 9(4): 5-15.

Madiega, T. (2020) *Digital sovereignty for Europe*, Brussels: European Parliamentary Research Service. https://www.europarl.europa.eu/thinktank/de/document/EPRS_BRI(2020)651992

Mathews, J. A., and Cho, T.-S. (2000): *Tiger Technology: The Creation of a Semiconductor Industry in East Asia*, Cambridge: Cambridge University Press.

Miailhe, N. (2018) 'The geopolitics of artificial intelligence: The return of empires?', *Politique étrangère*, (3): 105-17.

Miller, J., Keohane, D. and Kana, I. (2021) Car manufacturing hit by global semiconductor shortage. Financial Times, 08 January 2021. https://www.ft.com/content/e264fd41-7ee9-4fba-be3c-21446298efd9.

Mueller, M. (2020) 'Against Sovereignty in Cyberspace', *International Studies Review*, 22(4): 779-801.

Nye, Joseph S. 2014. 'The Regime Complex for Managing Global Cyber Activities.' *Global Commission on Internet Governance Paper Series*, No. 1. http://nrs.harvard.edu/urn-3:HUL.InstRepos:12308565.

Nyman, J. and Burke, A. (eds.) (2016) *Ethical Security Studies: A New Research Agenda*, Abingdon: Routledge.

Pasquale, F. (2016) 'Two Narratives of Platform Capitalism', *Yale Law & Policy Review*, 35: 309-19.

Pearson, G.L. and Brattain, W. H. (1955) 'History of Semiconductor Research', *Proceedings of the IRE* 43(12): 1794–1806.

Peterson, J. and Sharp, M. (1998) *Technology Policy in the European Union*, Houndmills: Macmillan.

Poell, T., Nieborg, D. and van Dijck, J. (2019) 'Platformisation', *Internet Policy Review*, 8(4).

Pohle, J. and Thiel, T. (2019) 'Digitale Vernetzung und Souveränität: Genealogie eines Spannungsverhältnisses', in I. Borucki and W. J. Schünemann (eds), *Internet und Staat: Perspektiven auf eine komplizierte Beziehung*, Baden-Baden, Nomos. pp. 57-80.

Pohle, J. and Thiel, T. (2020) 'Digital sovereignty', *Internet Policy Review*, 9(4).

Popiel, P. (2022) 'Regulating datafication and platformization: Policy silos and tradeoffs in international platform inquiries', *Policy & Internet*, 14(1): 28-46.

Roberts, H., Cowls, J., Casolari, F., Morley, J., Taddeo, M. and Floridi, L. (2021) 'Safeguarding European Values with Digital Sovereignty: An Analysis of Statements and Policies', *Internet Policy Review*, 10(3).

Rone, J. (2021) 'The return of the state? Power and Legitimacy Challenges to EU's Regulation of Online Disinformation', in B. Haggart, N. Tusikov and J. A. Scholte (eds), *Power and Authority in Internet Governance: Return of the State*, Routledge. pp. 171-94.

Ruohonen, J. (2021) 'The Treachery of Images in the Digital Sovereignty Debate', *Minds and Machines*, 31: 439-56.

Sahin, K. and Barker, T. (2021) *Europe's Capacity to Act in the Global Tech Race: Charting a Path for Europe in Times of Major Technological Disruption*, Berlin: German Council on Foreign Relations.

Schneider, N. (2022) 'Governable Stacks against Digital Colonialism', *triple C: Journal for a Global Sustainable Information Society*, 20(1): 19-36.

Strategic Forum (2019): Strengthening Strategic Value Chains for a future-ready EU Industry. Report of the Strategic Forum for Important Projects of Common European Interest. https://ec.europa.eu/docsroom/documents/37824.

Striphas, T. (2015) 'Algorithmic culture', *European Journal of Cultural Studies*, 18(4-5): 395-412.

van Dijck, J. (2020) 'Governing digital societies: Private platforms, public values', *Computer Law & Security Review*, 36.

von der Leyen. Ursula (2022): Statement by President von der Leyen on the European Chips Act. https://ec.europa.eu/commission/presscorner/detail/en/statement_22_866

Zeng, J., Stevens, T. and Chen, Y. (2017) 'China's Solution to Global Cyber Governance: Unpacking the Domestic Discourse of "Internet Sovereignty"', *Politics & Policy*, 45(3): 432-64.

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